



(AUTONOMOUS)

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PAPER PRESENTATION

Artificial Intelligence & Machine Learning



Introduction



- Artificial Intelligence (AI) and Machine Learning (ML) are transforming the way we interact with technology.
- AI focuses on creating systems that can simulate human intelligence, while ML enables systems to learn from data and improve performance over time.

Importance of AI & ML



- Automation of repetitive tasks
- Improved decision-making using data
- Enhanced user experiences (recommendations, personalization)
- Drives innovation in healthcare, finance, and robotics

Applications



- Virtual Assistants (Siri, Alexa)
- Self-Driving Cars
- Predictive Analytics in Business
- Image & Speech Recognition
- Healthcare Diagnosis & Drug Discovery

Challenges

- Data Privacy and Security Issues
- High Computational Costs
- Lack of Transparency (Black Box Problem)
- Need for Skilled Professionals

Future Scope

- AI & ML will continue to revolutionize industries by enabling advanced automation, real-time analytics, and personalized solutions. Integration with technologies like IoT, Blockchain, and Quantum Computing will open new opportunities.

Trends in AI & ML

- Generative AI (ChatGPT, DALL-E)
- Edge AI for IoT devices
- Explainable AI (XAI)
- AI in Cybersecurity
- Ethical & Responsible AI

AI & ML in Business

- Customer Segmentation
- Demand Forecasting
- Chatbots & Virtual Agents
- Fraud Detection
- Supply Chain Optimization

AI & ML in Education

- Personalized Learning Platforms
- Automated Grading Systems
- Virtual Tutors
- AI-based Career Guidance

AI & ML in Healthcare

- Disease Prediction
- Robotic Surgery Assistance
- Drug Discovery using AI
- Remote Patient Monitoring

AI & ML Tools

- Frameworks: TensorFlow, PyTorch
- Platforms: Google Vertex AI, AWS SageMaker
- Languages: Python, R

Data in AI & ML

Data is the backbone of ML models.

Types:

- Structured Data (Databases)
- Unstructured Data (Images, Text)
- Semi-structured Data (JSON, XML)

Data Preprocessing: Cleaning, Normalization,
Feature Engineering

Conclusion

- AI and ML are at the core of technological innovation. By adopting these technologies, businesses and society can unlock smarter solutions, better decision-making, and a more connected future.

THANK YOU